IV Assignment 3 Design Summary

Group 2

Base Map:

All three maps use the CartoDB base map, which is a simple map that contains very few place names but does include all road and suburb names, as well as some geographic features such as rivers and the ocean. This base map was chosen because we wanted to highlight the landmarks and other map information; thus, we chose a simple base map with minimal distracting features.
The boundary for the City of Melbourne is drawn over the base map so users can see the limits of our tool. The sidebar for the filtering options is positioned on the left for all maps for consistency, and the map itself is cantered on the screen for ease of use. All three maps are the exact same size, although the exact zoom level is not transferred between tabs.

All maps begin with all landmarks placed on the map. The dataset used here was a landmarks dataset from the City of Melbourne; however, we chose to filter out schools, residential accommodation, and vacant land as we believed these would not be useful or interesting places to visit for the average tourist. These landmarks are marked with circles on the maps, while map-specific information (such as train stations for the public transport maps) is represented by markers. This creates visual hierarchy for the user and communicates that the circle/dot landmarks are less important than the markers used for map-specific information. Once a landmark is selected, a circle around it appears showing the radius chosen from the sidebar, and the map-specific information contained within that radius also appears. It would be incredibly overwhelming and difficult to navigate if every single public transport stop were displayed on the map from the start, for example. Thus, we ensure that the important information is presented in a clean and readable manner.

All landmarks can be filtered by type, which is useful if a user is interested in looking for theatres, for example. All landmarks of the same type also have the same colour, and while some types share colours, they are usually of the same overall theme. Although the colours were somewhat arbitrarily decided, they have been chosen to not overlap with other information that can appear on the maps and generally follow a colour that most people would associate with that type. For instance, outdoor recreation facilities are all dark green, while medical/health-related landmarks are red.
The user is also able to search for any landmark in the dataset directly without interacting with the map. While the aim of the map and type filter is to aid the user in finding a landmark, we also provide the search function in case they already know what they are looking for. When a user selects a landmark on any map, all others are removed from the map until the user resets it, reducing clutter and allowing focus to be placed on the selected landmark. Finally, the user is able to select multiple landmarks at once. Since most tourists are likely interested in visiting multiple locations and may be using this tool to help plan their trip, it would be very frustrating if they could only see one landmark at a time.

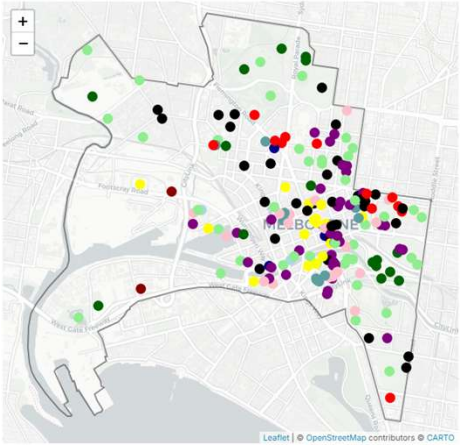
Public Transport Tab:

While making the public transport map, we aimed to create something not easily accessible through popular existing resources. For example, if a user wanted to find directions to a landmark or between landmarks, they would use something such as Google Maps, and if they were looking to find a timetable for a train service, they would use the PTV app. In our visualisation, we provide tools that would not be possible with either of those apps.

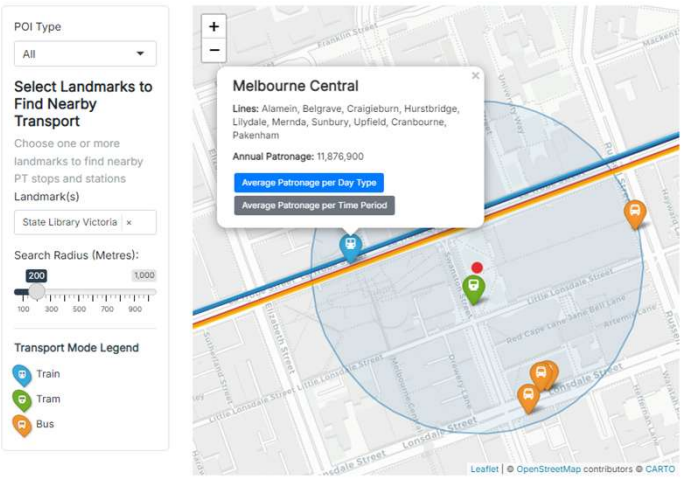
The datasets for the stops and lines were sourced from Transport Victoria. On the map, the colour of tram stops is lime green, bus stops are orange, and train stations are light blue. These colours correspond to the colours of the transport modes, and furthermore, all train lines have their associated colour as seen on platform signage and other PTV information displays (e.g., the Mernda line is red, the Upfield line is yellow).

When the user clicks on a landmark, the system will find all of the nearest tram and bus stops and all train stations within a certain radius. This radius can be adjusted using a slider on the sidebar, and although an app such as Google Maps will show all of the stops near a landmark, it is unable to filter for a specific distance. After clicking on a landmark, if the user clicks on one of the transport stops, all lines going through that stop will appear on the map. These lines will also extend beyond the City of Melbourne bounds to wherever they terminate, showing the full trip possible. This sort of feature doesn't really exist on other apps. While most will find the nearest path to another destination, they don't show all the possible options to other landmarks or find the best path to travel between multiple places.

After clicking on the stops/stations, a pop-up will also appear showing all routes/lines that can be caught there. For stations specifically, the user will also be able to see two different bar charts for patronage statistics. One shows the average patronage across different day types, and the other shows the average patronage across different time periods throughout the day. This data is not available on common transport apps and can aid the user in seeing what times are busy and when it is best to take public transport, allowing them to plan their trip around this information.



Base Map with Boundary and Landmarks



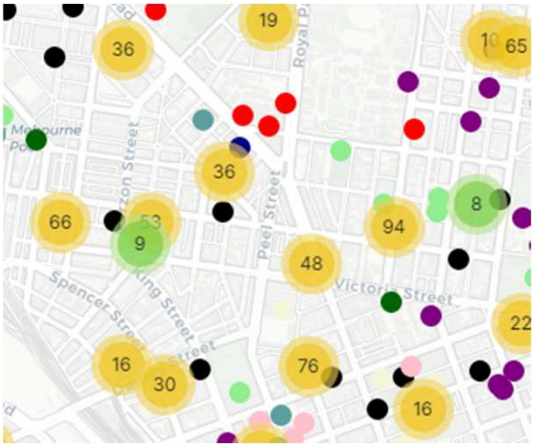
Public Transport Map: On Click Station Lines and Information

On-Street Parking Tab:

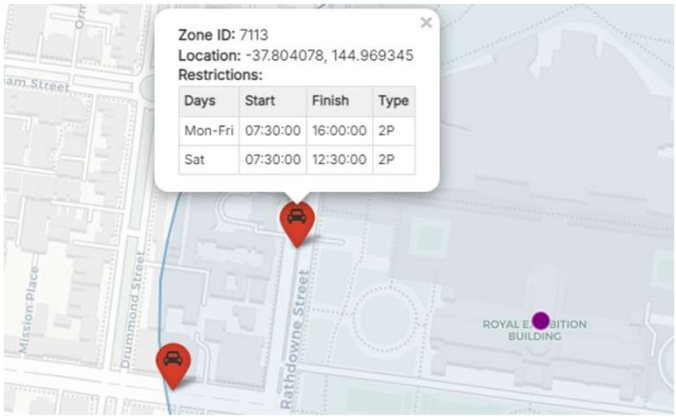
While public transport is used by most tourists, some people still prefer to drive, and thus we also provide a visualisation for on-street parking. The data used in this tab is sourced directly using the City of Melbourne API, providing a live view of all on-street parking places available and refreshing on each load of the app.

Similar to the public transport map, this uses the same base map and landmark colour dots, along with the landmark sorting/search and radius slider. This works in a similar way to the public transport map, where the user can click on a landmark and see all the available parking locations nearby. The map includes all of the individual parking spots as markers but in a clustered view, with them unclustering into individual markers when zoomed in but appearing as clustered groups when zoomed out to reduce clutter and improve visual clarity.

When zoomed in or clicked on, the clusters separate into individual markers. Red markers represent parking spaces with a specific zone ID, and upon clicking on them, the user will see a table with all the parking restrictions for that space at different times. Orange markers represent parking spots where the zone is unknown; although it is an available spot, the user is sufficiently warned that further information is unavailable.



On-Street Parking Map: Zoomed Out Clusters



On-Street Parking Map: Selected Marker and On Click Information

Pedestrian Count Tab:

The pedestrian traffic tab is designed to help users explore how people move around the City of Melbourne and which landmarks are most attractive to people. It brings together three main datasets: landmark locations, pedestrian sensor counts, and sensor coordinates, in order to show how pedestrian activity connects to key landmarks. All datasets here are sourced from the City of Melbourne, and the interface includes three interactive views: Heatmap View, Ranking View, and Trend View.

The Heatmap View shows pedestrian density across Melbourne using colour intensity. For the visual design, the heatmap gradient highlights high-traffic areas from navy to red, with red representing the higher-traffic areas. This map uses the same sidebar and base as the previous maps, and when a user clicks on a landmark, they are able to see the exact sensors near the landmark and the exact number of pedestrians detected by the sensor.

The Ranking View displays the top 15 most visited landmarks as a bar chart, making it easy to compare which areas attract the most foot traffic. In addition, the user is able to filter by landmark type, and the colours used in this chart correspond to the same colours that the landmark types have on the maps. The Trend View produces a line graph visualising how pedestrian activity changes over time. Once again, the user is able to filter by landmark type, and combined, these two sections can aid users in seeing patterns such as busy weekends or seasonal spikes and allow them to plan their trips accordingly. Combining these three views together, we provide both spatial and temporal understanding of city movement patterns in a single, easy-to-use interface. This structure ensures that the visual design matches the data meaningfully and draws the user's attention to key insights first.

Pedestrian Count Map: Individual Sensor Information

Security Tab:

The "Security View" interface is designed to explore recorded crime data for the Melbourne Local Government Area (LGA) for the year ending June 2025. Its primary purpose is to transform raw statistical tables into actionable insights. The interface's core functions include a high-level spatial overview via an interactive choropleth map, contextual overlays of landmarks, and a multi-tab panel that presents city-wide aggregated statistics.

The interface's logic is built around a series of coordinated reactive components. On startup, the application preloads and processes several data tables from the source Excel file. The "Interactive Map" tab features a Leaflet map, with its data driven by a reactive expression that filters suburb-level offence data based on the user's selection from the "Crime Category" dropdown. This allows the map to function as a dynamic choropleth, colouring suburb polygons on a sequential scale from light yellow to red to represent offence volume. A key interaction pattern is the "drill-down", where clicking a suburb polygon reveals a pop-up with "Categories Details" and "Data Details" buttons. These buttons use JavaScript to send the suburb's name back to the server, which then displays a modal dialogue with filtered charts and tables, providing details-on-demand.

The entire "Security View" workflow is a direct implementation of a progressive disclosure strategy, designed to manage cognitive load. The "Overview" tab provides the highest-level summary, presenting key performance indicators such as total offences and crime rate. The "Interactive Map" allows for spatial exploration and filtering. The modal dialogues provide the most granular detail, but only when the user explicitly requests it. This prevents information overload and maintains the user's spatial context, as they are not forced to navigate away from the map to see the underlying data. This approach provides a clear path for analysis: overview first, then zoom and filter, followed by details-on-demand.

The design relies on appropriate visual encodings for each data type. The main choropleth map uses a sequential colour palette, which leverages pre-attentive processing to make crime "hotspots" immediately apparent. This is supported by other visualisations in the coordinated tabs, such as a Plotly sunburst chart in the "Overview" tab. This sunburst effectively displays the hierarchical nature of offence location data (Division to Subdivision to Group). Standard bar and pie charts are used in the "Investigation Status" and "Drug Offences" tabs to clearly present categorical and part-to-whole comparisons, ensuring the chosen visualisation is fit for purpose.

IV Assignment 3 Pattern and Use Case Summary

Group 2

Public Transport Tab:

The best use of this dashboard would be for tourists who are planning to use public transport to travel around the city and have multiple places they want to visit. While Google Maps can create directions to go from one landmark to another, it often provides very complex travel plans and may not suggest the best route order. For example, if a user wants to plan their day and visit Federation Square, City Baths, and Melbourne Zoo, they could select all three on the public transport map and, by clicking on the stops, find out that one tram trip is all that is necessary between City Baths and Federation Square, and only one train trip is necessary between Federation Square and Melbourne Zoo (using a 500 m radius). This provides a quick and simple plan to follow, with only one mode of transport needed for each trip, and this is very useful for tourists as they are likely unfamiliar with the public transport system. Google Maps can be notorious for generating very overcomplicated travel paths in cities with numerous transport options.

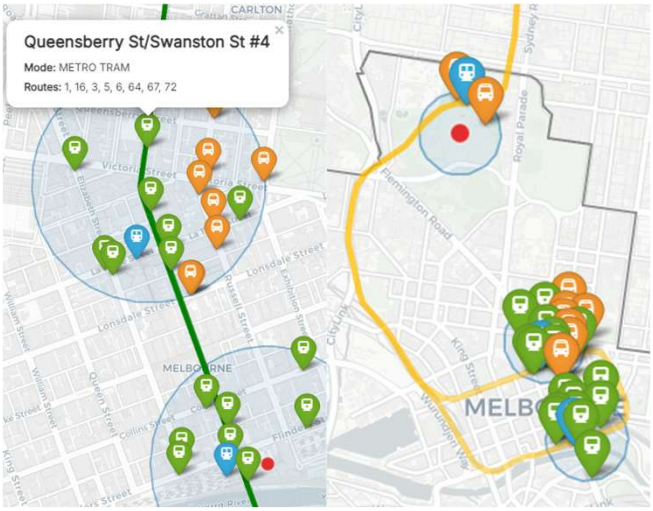
Station patronage information may also help tourists who want to avoid large crowds. For example, if a user wants to visit the State Library in the afternoon, they will find that the nearest station is Melbourne Central. They can then look at the charts and see that the station is quietest on Saturdays or Sundays, and thus, over a week-long trip, they may decide to spend part of their weekend visiting this, as coming on a weekday afternoon could be much more stressful due to the much larger crowds.

On-Street Parking Tab:

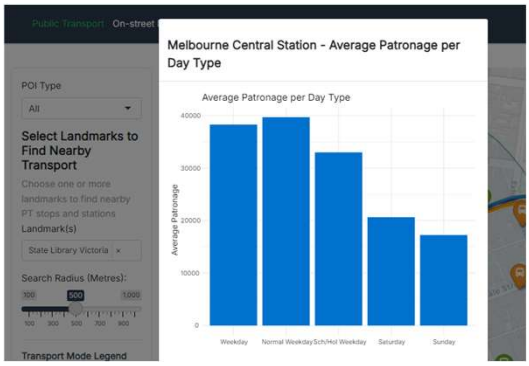
As not every tourist will come to Melbourne intending to use public transport to move around the city, tourists who drive will likely find this visualisation very useful, as searching for parking in and around the Melbourne CBD is notoriously difficult due to how busy it is. For users driving in, they may use other tools such as Google Maps for directions; however, when they get close to the destination, they can use our map to find a nearby parking space.

This tool is not particularly useful for solo travellers, as they would not be allowed to use it while driving. However, it could be very useful for a passenger to check where to park when they get close to their destination. If there is only one parking space, then it may be risky to still go for it, as it could easily be taken by someone else. However, that is where the cluster view can also help with seeing many available spaces in one area, as the user can decide to drive to an area with many open spaces, maximising their chances of finding a place to park.

Furthermore, this part of the app is also likely useful for residents of Melbourne and nearby areas. While the public transport map is not as useful for locals who are already familiar with the network, anyone who is driving into Melbourne and looking for parking can find this visualisation useful, especially as it updates available parking locations in real-time upon app launch.



Public Transport Map: Journey Planning Example



Public Transport Map: Patronage Metrics Example

On-Street Parking Map: Clusters and Markers Example

Pedestrian Tab:

The dashboard is designed especially for tourists and visitors to Melbourne. It helps them quickly understand which areas of the city are hotspots, what kind of places attract the most people, and when those places are most active.

For example, a first-time visitor could start with the Heatmap View to get an overview of where the most vibrant parts of Melbourne are. The red and orange zones highlight popular pedestrian areas such as Swanston St and Bourke St. By clicking on a landmark, users can see information about the landmark and the average pedestrian counts. Next, they can move to the Ranking View, which lists the top 15 landmarks by pedestrian volume.

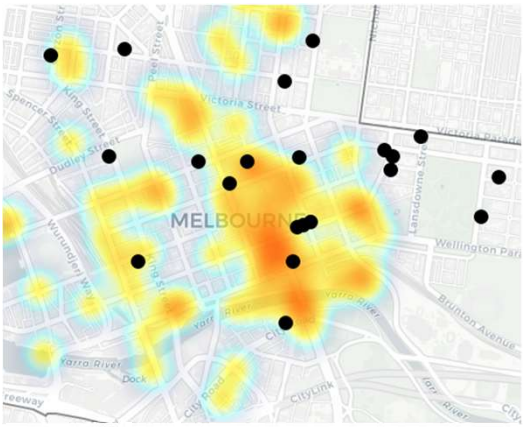
Without a travel guide, tourists can still find out the must-visit attractions in Melbourne. Finally, the Trend View helps visitors plan when to go. For instance, they can see that cultural and recreational areas are busiest on weekends, while retail zones peak on weekday afternoons. This makes it easier for tourists to schedule their visit. They can avoid the heaviest crowds or choose the best time for sightseeing and photography.

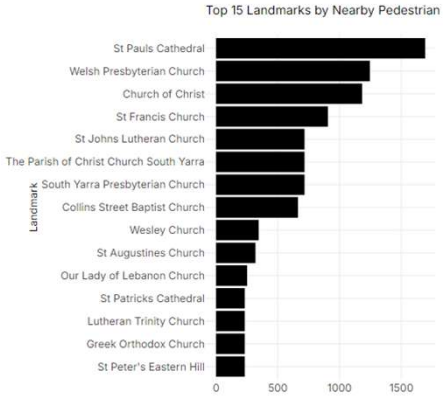
The interactive design turns a large amount of sensor data into an intuitive visual representation that tells a story about the city's rhythm. Tourists can find not only where people gather but also understand when and why those areas are most appealing.

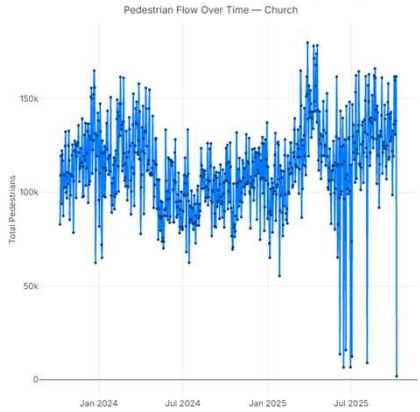
Security Tab:

This interface is designed to serve multiple user personas by allowing them to discover patterns at different scales. A prospective resident, Anna, can use the tool to compare the safety profiles of suburbs she is considering. Upon loading the "Security View", she immediately sees a clear spatial pattern on the map: the central, high-density areas of "Melbourne" (CBD) and "Southbank" are shaded dark red, indicating a much higher volume of total offences. By changing the "Crime Category" filter to "Property & Deception", she can update the map and confirm this pattern holds for her specific concern. She then clicks on the "Carlton" and "Southbank" polygons and uses the "Categories Details" modal to view bar charts, providing quantitative evidence to support her visual assessment. The tool helped Anna by providing a rapid spatial overview, interactive filtering, and a seamless drill-down workflow.

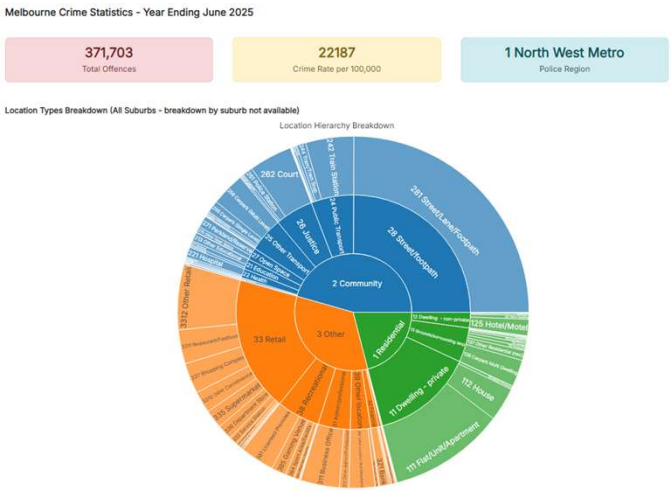
Similarly, an urban policy adviser, David, can use the tool to analyse public space safety. He navigates to the "Overview" tab and examines the "Location Types Breakdown" sunburst chart. He quickly discovers that "Community Locations" is a major category and drills in to find that "Street/Footpath" and "Licensed Premises" are the dominant subcategories. To confirm this, he filters the "Interactive Map" to "Public Order" offences and sees the "Melbourne" and "Southbank" suburbs highlighted, confirming his hypothesis about entertainment and high-traffic zones. Finally, he visits the "Investigation Status" tab to see the proportion of unsolved cases, giving him a complete picture for his report. The tool's coordinated multiple views were critical, allowing him to synthesise insights from spatial, hierarchical, and categorical data to build an evidence-based case.



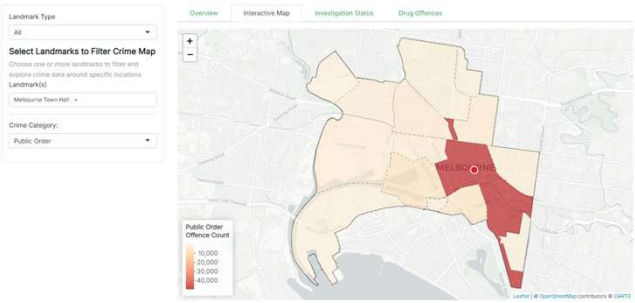




Pedestrian Counts Tab: In Depth Viewing Example of Churches



Security Tab: Overview Location Examination Example



Security Tab: Interactive Map Filtering Example

Data Sources

Boundary:
https://data.melbourne.vic.gov.au/explore/dataset/municipal-boundary/information/

Landmarks:
https://data.melbourne.vic.gov.au/explore/dataset/landmarks-and-places-of-interest-including-schools-theatres-health-services-spor/information/

Public Transport Stops:
https://opendata.transport.vic.gov.au/dataset/public-transport-lines-and-stops

Public Transport Lines:
https://opendata.transport.vic.gov.au/dataset/public-transport-lines-and-stops

Train Station Patronage:
https://opendata.transport.vic.gov.au/dataset/annual-metropolitan-train-station-patronage-station-entries

On-Street Parking Bays:
https://data.melbourne.vic.gov.au/explore/dataset/on-street-parking-bays/api/

Parking Zones:
https://data.melbourne.vic.gov.au/explore/dataset/parking-zones-linked-to-street-segments/api/

Parking Zone Sign Plates:
https://data.melbourne.vic.gov.au/explore/dataset/sign-plates-located-in-each-parking-zone/api/

Pedestrian Counts:
https://data.melbourne.vic.gov.au/explore/dataset/pedestrian-counting-system-monthly-counts-per-hour/information/

Pedestrian Sensor Locations:
https://researchdata.edu.au/pedestrian-counting-system-sensor-locations/2296347

Crime Dataset:
https://discover.data.vic.gov.au/dataset/data-tables-recorded-offences/resource/4c1c7d2a-3326-493c-8b5f-ee35bad60499

Group Contribution

Name	Contribution	Quality	Percentage
Connor Liu	Code: Public Transport Tab Report: Base Map, Public Transport Tab, Parking Tab Video: Introduction and Public Transport Tab	Contributed heavily to team discussions, always completed work on time	25%
Zhuofei Yan	Code: Unified base map with filters, Parking Tab, Crime Tab Map Video: Parking Tab	Contributed heavily to team discussions, always completed work on time	25%
Shu-Han Yang	Code: Pedestrian Tab, Tableau Integration Report: Pedestrian Tab Video: Pedestrian Tab	Contributed heavily to team discussions, always completed work on time	25%
Anthony He	Code: Crime Tab Charts Report: Crime Tab Video: Crime Tab	Contributed heavily to team discussions, always completed work on time	25%